

BRD Frontend Design — Cross-Reference Gap Analysis

Date: April 2026
Project: Smart URL Shortener Platform
Purpose: Identify missing APIs, missing screens, and mismatches between BRD and Frontend Design

1. API Inventory (from BRD)

Auth Module (4 APIs)

#	Method	Endpoint	Frontend Screen(s)
A1	POST	/api/v1/auth/register	Register Page
A2	POST	/api/v1/auth/login	Login Page
A3	POST	/api/v1/auth/refresh	Global (silent token refresh)
A4	POST	/api/v1/auth/logout	Navbar → Logout

URL Module (5 APIs)

#	Method	Endpoint	Frontend Screen(s)
U1	POST	/api/v1/urls	Landing Page, Dashboard, Create URL
U2	GET	/api/v1/urls	Dashboard, My URLs
U3	GET	/api/v1/urls/{id}	URL Detail, Edit URL
U4	PUT	/api/v1/urls/{id}	Edit URL
U5	DELETE	/api/v1/urls/{id}	My URLs, URL Detail

Redirection (1 API)

#	Method	Endpoint	Frontend Screen(s)
R1	GET	/ {shortCode}	External (redirect)

Analytics (2 APIs)

#	Method	Endpoint	Frontend Screen(s)
N1	GET	/api/v1/analytics/{urlId}	URL Detail + Analytics
N2	GET	/api/v1/analytics/dashboard	Dashboard

Admin (6 APIs)

#	Method	Endpoint	Frontend Screen(s)
M1	GET	/api/v1/admin/users	Admin Users Management
M2	PUT	/api/v1/admin/users/{id}/block	Admin Users Management
M3	GET	/api/v1/admin/urls	Admin URLs Management
M4	PUT	/api/v1/admin/urls/{id}/disable	Admin URLs Management
M5	DELETE	/api/v1/admin/urls/{id}	Admin URLs Management
M6	GET	/api/v1/admin/dashboard	Admin Dashboard

Total BRD APIs: 18

2. Screen Inventory (from Frontend Design)

Public Screens

Route	Screen	APIs Used
/	Landing Page	U1 (POST shorten), R1 (redirect)
/login	Login Page	A2
/register	Register Page	A1

User Screens

Route	Screen	APIs Used
/dashboard	Dashboard	U2, N2, U1
/urls	My URLs	U2, U5, U4
/urls/new	Create URL	U1
/urls/:id/edit	Edit URL	U3, U4
/urls/:id/analytics	URL Analytics	U3, N1
/profile	User Profile	NONE DEFINED

Admin Screens

Route	Screen	APIs Used
/admin/dashboard	Admin Dashboard	M6, M1, M3
/admin/users	Admin Users	M1, M2

Table 8 – continued

Route	Screen	APIs Used
/admin/urls	Admin URLs	M3, M4, M5

3. CRITICAL GAPS — Missing APIs (Frontend needs, BRD missing)

3.1 User Profile APIs

The Frontend Design has a full **Profile Page** (/profile) with:

- View/Edit profile info (full name, username, email, phone)
- Change password
- View active sessions
- Revoke sessions

Missing BRD APIs:

Needed API	Method	Endpoint	Purpose
Get Current User	GET	/api/v1/users/me	Load profile data
Update Profile	PUT	/api/v1/users/me	Save profile changes
Change Password	PUT	/api/v1/users/me/ password	Update password
Get Sessions	GET	/api/v1/users/me/ sessions	List active sessions
Revoke Session	DELETE	/api/v1/users/me/ sessions/{id}	Logout from device
Delete Account	DELETE	/api/v1/users/me	Account deletion
Upload Avatar	POST	/api/v1/users/me/ avatar	Profile photo

Impact: HIGH — Profile page is fully designed but has zero backend support.

3.2 Alias Availability Check API

The Frontend Design shows **real-time alias availability checking** with debounce:

Typing... → Debounce 500ms → API check →  Available /  Taken

Missing BRD API:

Needed API	Method	Endpoint	Purpose
Check Alias	GET	/api/v1/urls/ check-alias? alias={alias}	Validate alias availability before submission

Impact: MEDIUM — Without this, users only find out alias is taken after form submission.

3.3 Raw Click Logs API

The Frontend Design shows a “**Recent Clicks Log**” table on the Analytics page with individual click records:

Time IP Address Browser Device Country
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The BRD only defines aggregated analytics (GET /api/v1/analytics/{urlId}) which returns totals/charts.

Missing BRD API:

Needed API	Method	Endpoint	Purpose
Get Click Logs	GET	/api/v1/analytics/ {urlId}/clicks	Paginated raw click records

Impact: MEDIUM — The “Recent Clicks” table cannot be populated without this.

3.4 Public Stats API

The Frontend Landing Page shows a **Stats Banner**:

1M+ URLs Shortened 50M+ Clicks Tracked 99.9% Uptime 10K+ Users
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Missing BRD API:

Needed API	Method	Endpoint	Purpose
Public Stats	GET	/api/v1/public/stats	Platform-wide public statistics

Impact: LOW — Can be hardcoded initially, but needs backend for real data.

3.5 Password Reset / Forgot Password APIs

The Frontend Login Page has a “**Forgot password?**” link.

Missing BRD APIs:

Needed API	Method	Endpoint	Purpose
Request Reset	POST	/api/v1/auth/ forgot-password	Send reset email
Reset Password	POST	/api/v1/auth/ reset-password	Confirm reset with token

Impact: MEDIUM — Common real-world feature, but can be deferred to Phase 7.

3.6 Email Verification APIs

The BRD mentions “Email verification support” and the Frontend Register has terms checkbox, but no verification flow APIs exist.

Missing BRD APIs:

Needed API	Method	Endpoint	Purpose
Resend Verification	POST	/api/v1/auth/ verify-email/resend	Resend verification email
Verify Email	GET	/api/v1/auth/ verify-email? token={token}	Confirm email via token

Impact: LOW — Can be deferred; BRD mentions it as a feature but no endpoints.

3.7 QR Code Generation API

The BRD mentions “QR code generation for short URLs” and Frontend shows QR codes, but no dedicated API.

Missing BRD API:

Needed API	Method	Endpoint	Purpose
Get QR Code	GET	/api/v1/urls/{id}/qr	Generate/download QR code

Impact: LOW — Can be generated client-side or as part of URL response.

3.8 Bulk Operations APIs

The Frontend Admin pages show “**Block Selected**” and “**Delete Selected**” bulk action buttons.

Missing BRD APIs:

Needed API	Method	Endpoint	Purpose
Bulk Block Users	PUT	/api/v1/admin/users/ bulk/block	Block multiple users
Bulk Delete URLs	DELETE	/api/v1/admin/urls/ bulk	Delete multiple URLs
Bulk Disable URLs	PUT	/api/v1/admin/urls/ bulk/disable	Disable multiple URLs

Impact: LOW — Can loop single APIs, but inefficient for large selections.

3.9 Admin User Detail / Role Update APIs

The Frontend Admin Users page shows actions: “**View Profile** | **Edit Role** | **Block/Unblock** | **Delete Account**”

Missing BRD APIs:

Needed API	Method	Endpoint	Purpose
Get User Detail	GET	/api/v1/admin/users/ {id}	View full user profile
Update User Role	PUT	/api/v1/admin/users/ {id}/role	Change USER ADMIN
Delete User	DELETE	/api/v1/admin/users/ {id}	Hard delete user

Impact: MEDIUM — “Edit Role” and “Delete Account” actions have no backend.

3.10 URL Password Update API

The Frontend URL Detail page shows a “**Password**” action button to add/change link password.

Missing BRD API:

Needed API	Method	Endpoint	Purpose
Update URL Password	PUT	/api/v1/urls/{id}/ password	Add/change/remove link password

Impact: LOW — Link password is a Phase 6 advanced feature.

4. 🟡 MISSING SCREENS (BRD mentions, Frontend missing)

4.1 Email Verification Page

The BRD mentions “Email verification support” but Frontend has no verification success/failure page.

Missing Screen: `/verify-email?token=xxx` — Show success/error message

4.2 Password Reset Pages

The BRD doesn't define APIs, but Frontend has “Forgot password?” link with no destination.

Missing Screens:

- `/forgot-password` — Enter email form
- `/reset-password?token=xxx` — New password form

4.3 URL Password Prompt Page

The BRD mentions “URL password-protected → Password prompt page” but Frontend has no such screen.

Missing Screen: `{shortCode}?password=true` — Password input before redirect

4.4 Error Pages

The BRD defines multiple redirect error states (404, 410, 403) but Frontend has no error pages.

Missing Screens:

- `/error/404` — Short code not found
- `/error/410` — URL expired
- `/error/403` — URL disabled

4.5 Bulk URL Upload Page

The BRD mentions “Bulk URL upload (CSV)” as an advanced feature but Frontend has no upload screen.

Missing Screen: `/urls/bulk-upload` — CSV upload with preview

4.6 Admin Settings Page

The Frontend Admin Sidebar has a “Settings” menu item (`/admin/settings`) but no corresponding screen design.

Missing Screen: `/admin/settings` — Platform configuration

5. 🟢 ORPHAN APIs (BRD defines, Frontend doesn't use)

API	Endpoint	Issue
POST /api/v1/auth/refresh	A3	Used silently by Axios interceptor, not tied to a visible screen. OK –background usage.
GET /{shortCode}	R1	External redirect, not a frontend screen. OK –backend-only.

Result: No true orphan APIs. All BRD APIs have frontend consumers.

6. ⚠️ MISMATCHES & AMBIGUITIES

6.1 Analytics Endpoint Mismatch

- **BRD:** GET /api/v1/analytics/url/{urlId} (Section 5.4, original doc) vs GET /api/v1/analytics/{urlId} (merged BRD)
- **Recommendation:** Standardize to GET /api/v1/analytics/{urlId}

6.2 Admin Dashboard Stats

- **BRD:** GET /api/v1/admin/dashboard returns platform analytics
- **Frontend:** Admin Dashboard shows “Total Users, Total URLs, Total Clicks, Suspicious URLs”
- **Gap:** The “Suspicious URLs” count needs a separate data source or filter on the admin dashboard API.

6.3 Guest Shorten on Landing Page

- **BRD:** Guest can create limited URLs (rate-limited, 5/day)
- **Frontend:** Landing Page allows shortening without login
- **Gap:** Need to clarify if guest-created URLs are stored in DB with `user_id = NULL` or require a temporary session. The DB schema shows `user_id BIGINT REFERENCES users(id)` which allows NULL, so this works.

6.4 URL Soft Delete

- **BRD:** DELETE /api/v1/urls/{id} says “Delete URL (soft delete)”
- **Frontend:** Shows “Delete” button with confirmation
- **Gap:** If soft delete, should the “My URLs” page show deleted URLs with a “Restore” option? Currently Frontend has no “Trash/Deleted” filter.

7. 📁 SUMMARY MATRIX

By Priority

Priority	Item	Type	Count
 HIGH	User Profile APIs	Missing API	7
 HIGH	Alias Availability API	Missing API	1
 MEDIUM	Raw Click Logs API	Missing API	1
 MEDIUM	Password Reset APIs	Missing API	2
 MEDIUM	Admin User Detail/ Role APIs	Missing API	3
 LOW	Public Stats API	Missing API	1
 LOW	Email Verification APIs	Missing API	2
 LOW	QR Code API	Missing API	1
 LOW	Bulk Operation APIs	Missing API	3
 LOW	URL Password Update API	Missing API	1
 MEDIUM	Error Pages (404/410/403)	Missing Screen	3
 MEDIUM	Password Reset Pages	Missing Screen	2
 LOW	Email Verification Page	Missing Screen	1
 LOW	URL Password Prompt	Missing Screen	1
 LOW	Bulk Upload Page	Missing Screen	1
 LOW	Admin Settings Page	Missing Screen	1

Total Missing APIs: 22

Total Missing Screens: 9

8. 💡 RECOMMENDATIONS

Immediate (Phase 2-3)

1. **Add User Profile APIs** — The profile page is fully designed and must have backend support
2. **Add Alias Check API** — Critical for UX during URL creation
3. **Standardize analytics endpoint** — Pick one path format

Short-term (Phase 4-5)

4. **Add Raw Click Logs API** — Required for the “Recent Clicks” table
5. **Add Password Reset APIs + Screens** — Complete the auth flow
6. **Add Admin User Detail/Role APIs** — Complete admin user management
7. **Add Error Pages** — 404, 410, 403 for redirect failures

Long-term (Phase 6-7)

8. **Add Email Verification APIs + Screen** — Production-ready auth
9. **Add Bulk Operations APIs** — For admin efficiency
10. **Add Public Stats API** — For landing page credibility
11. **Add QR Code API** — If not generating client-side

12. **Add URL Password Prompt Screen** — For protected links
13. **Add Bulk Upload Screen** — CSV upload feature

End of Gap Analysis